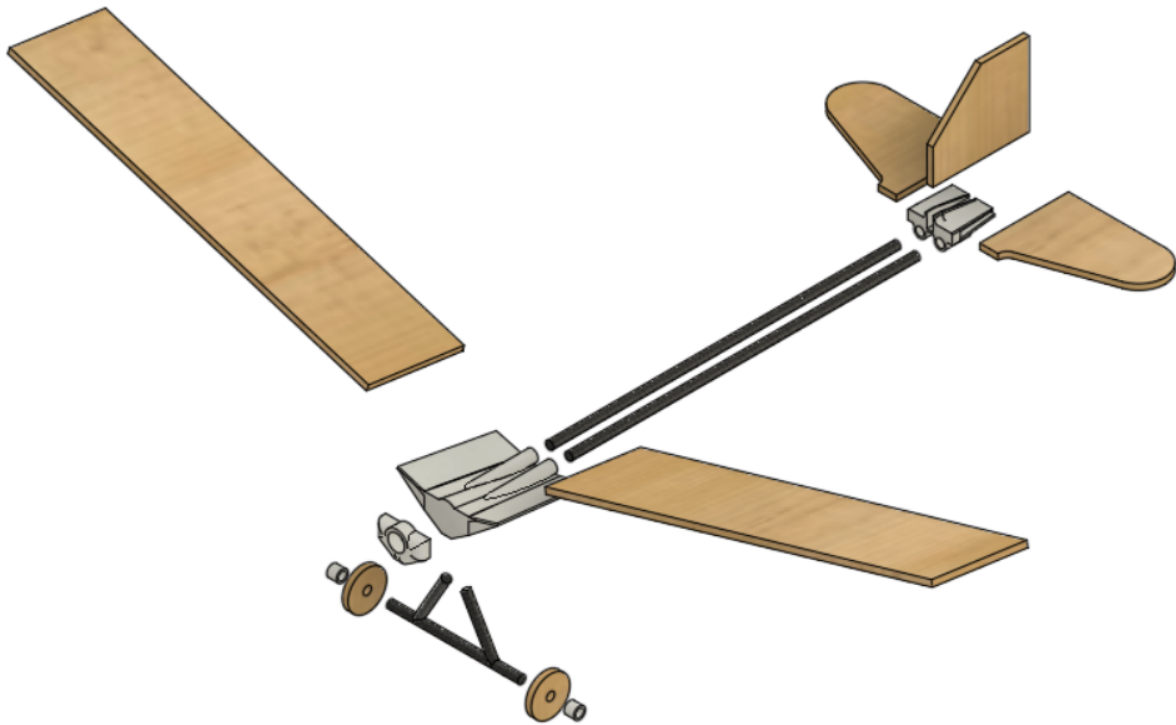


AER11002

Group Design Build and Test Project: Around the Pole Glider



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Group 6

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Introduction

Background:

Our task is to design and build a flying glider that can fly around a pole, using only the materials provided to us. We need to apply our design, manufacturing, and theoretical knowledge to optimise the glider's efficiency. We will use aerodynamic principles and workshop facilities to construct the model, and evaluate its performance by analysing the power needed for takeoff and sustained flight.

Objective:

1. To apply the theory learned in AER11002 to the design of a model glider.
2. To demonstrate the ability to design, analyse, and justify design choices through the creation of a round-the-pole flyer.
3. To ensure the glider can fly horizontally at the height of the tether for at least one circuit.
4. To build the model glider only using the available workshop facilities.
5. To evaluate the performance of the model glider by analysing the power required for takeoff and sustained flight.

Scope:

This report will include the following:

1. An overview of the research that went into the design of the glider
2. A detailed account of the design process, including any testing and design modifications.
3. A description of the materials, tools, and equipment used in the design and construction of the model glider.
4. A step-by-step guide to the assembly process, including any challenges encountered and how they were overcome.
5. A description of the testing process and the criteria used to evaluate the performance of the model glider.
6. A presentation of the results of the testing and a comparison of the results with similar model gliders.
7. A conclusion summarising the key findings of the project and its significance.

Design

Material Selection:

We opted for carbon fiber rods and 3D-printed parts to reduce weight and ensure precision. Balsa wood was chosen for the wheels, wings, and tailplane due to its lightweight nature and relative strength. Joining the parts is a concern, as tape and glue would add weight, so we aim to minimize their use by using superglue. Our decision to use carbon fiber rods instead of balsa wood for a fuselage is also based on their durability and toughness, reducing the risk of material failure during flight. Additionally, 3D-printed parts offer the advantage of being able to be manufactured into complex shapes with high accuracy, while balsa wood is the easiest to cut for the wheels.

Initial Design:

Originally, we planned to use a wooden fuselage glider like the AER11002 glider project, but we wanted a lighter aircraft. To do this, we focused on the heaviest part (the fuselage) and decided to use CFRP rods instead. This led us to consider how to attach the wings, landing gear, motor, and tailplane to the rods.

We knew we had 3D printing at our disposal, and could therefore make some custom parts to hold everything together. These parts were designed in Fusion:

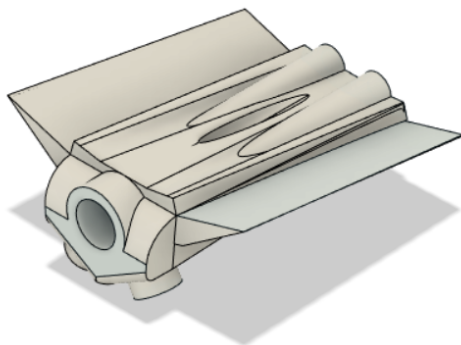


Figure 1.1

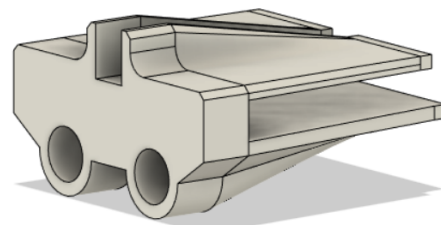


Figure 1.2

Figure 1.1 - The main body, which joins the wings using glue to the top surfaces, the motor in the front, landing gear underneath and fuselage rods in the back.

Figure 1.2 - The tail joint, which holds the tailplanes and the rudder orthogonally when slotted in, as well as having two holes in the front for the fuselage rods.

By using carbon rods for the fuselage we could avoid having to calculate the centre of mass of the plane until after it was built, as we could simply cut the rods down right before attaching the tail piece to give us our desired centre of mass. To optimise lift for the glider's wings, we chose a high angle of attack of 8° to generate the primary lift force. We also agreed on a dihedral angle to avoid collision with the floor in case the wing tip wire pulled on the wing. The worst-case scenario required a 15° angle on each wing since the wire and floor made a 30° angle. This is demonstrated in Figure 2.



Figure 2

The landing gear was going to be two carbon rods glued into a hole underneath the wing joint, with a groove sanded into the bottom so the axle can be glued flush and in place. The wheels would be laser cut with a clearance fit to allow them to spin, and 3D-printed stoppers would hold them in place like shown in Figure 3.

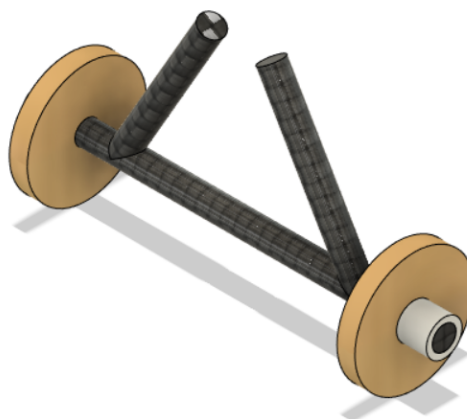


Figure 3

Revised Design:

We needed to print the parts such that there were no need for supports in the print. This involved slicing parts so they can be glued together after the print. Also, we decided to make the following changes to the designs:

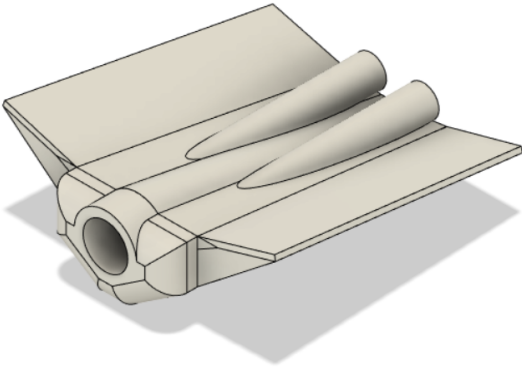


Figure 4.1

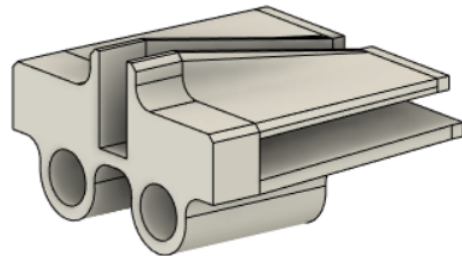


Figure 4.2

Figure 4.1 - We simplified the wing joint, taking out the holes for the motor wire and the landing gear, since we can drill and glue as needed.

Figure 4.2 - We added more stability for the rudder, making a split in the tailplanes, and opened the back of the fuselage holes for COG adjustment

Manufacture

Tools Used:

Laser Cutter

To produce our tail components, see Figure 5, the wings and the wheels, we utilised the laser cutter. With this method we ensured our pieces were cut accurately and to the correct size with as little waste as possible. Our first time using the laser cutter resulted in our wood catching fire so we adjusted the setting and the scale of the machine.

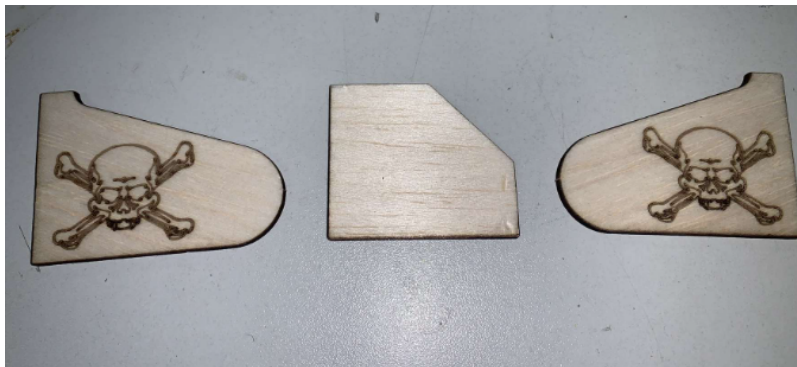


Figure 5

3D Printer

For our main sections of the plane we produced them using the 3D printer. See Figure 6, we printed the wing joiner, the motor holder, tail plane joiner and the rings for the wheels. In our first attempt there wasn't enough clearance for the carbon rods so we adjusted it in CAD to create a refined version.

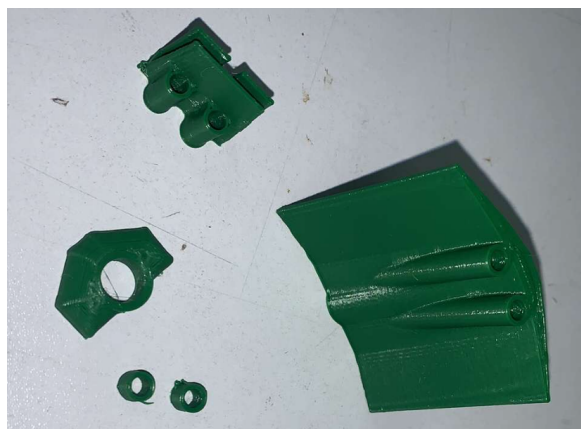


Figure 6

Hand tools

To cut our carbon rods we first used a scroll saw, however that caused the rods to snap and fray which could lead to in-flight issues, so we decided to use a hacksaw instead. Figure 7 shows us marking the carbon rods before we cut them. The tape made it easier to cut accurately with the saws.

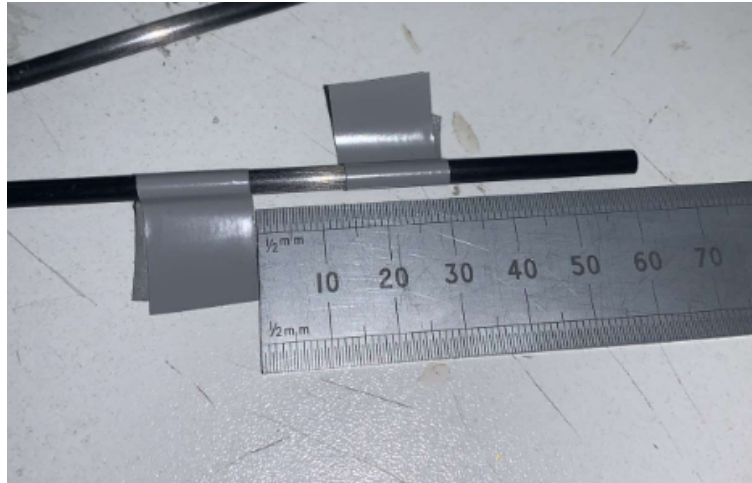


Figure 7

Pillar drill

We used a pillar drill to drill a hole into the main body to fit the motor better. We could now get the wires out and run them down the wings so that they could connect to the power. When using the drill the layers of the plastic began to fray and this made it more difficult to feed the wire through, however we would clear the debris out and the wire fed through as shown in Figure 8.

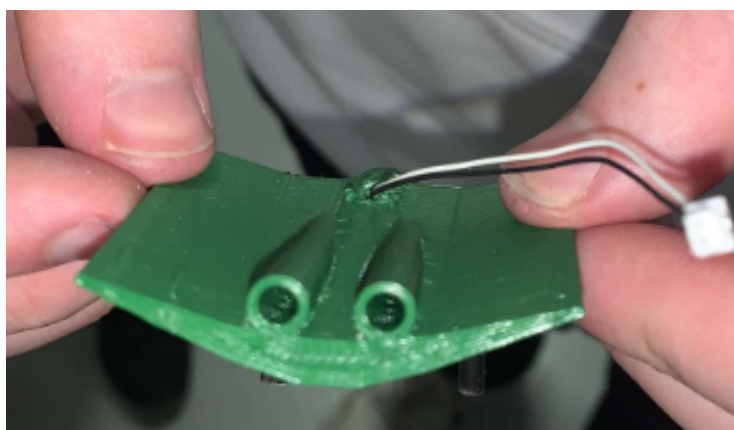


Figure 8

Assembly:

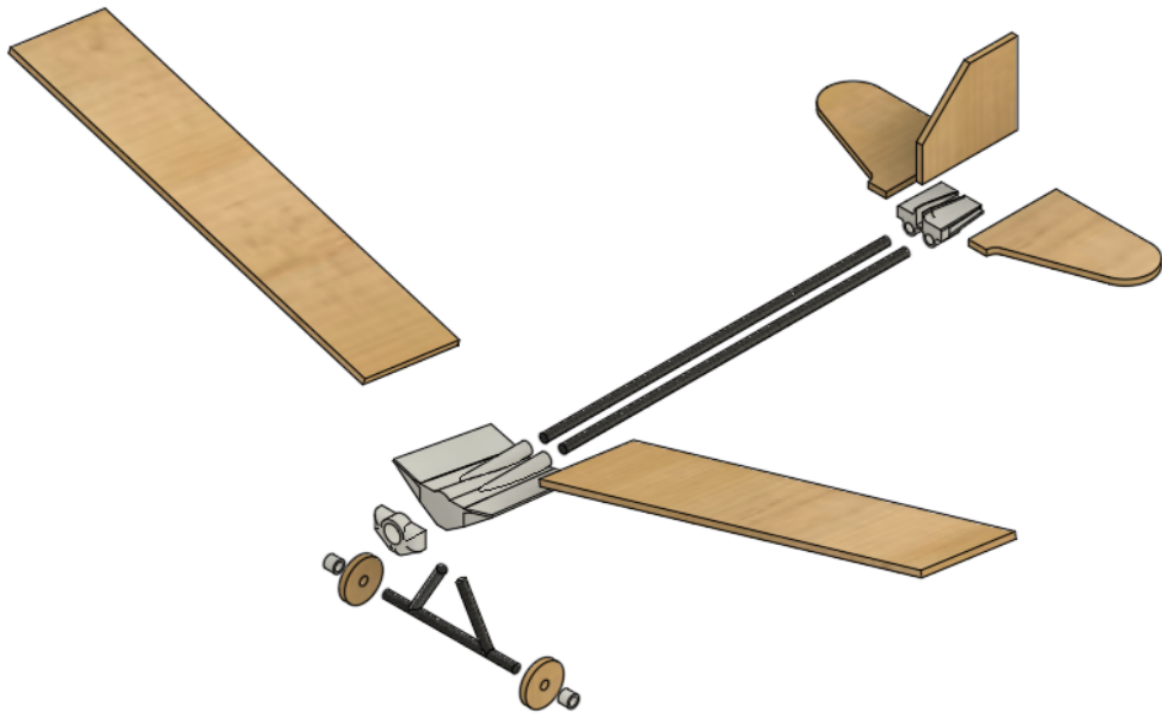


Figure 9 - The assembly plan

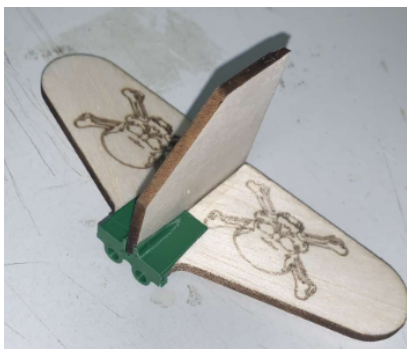


Figure 10

Tail Plane

The balsa wood fit perfectly into the tail joint. When we started to glue the joint snapped on the right side since the forces were applied parallel to the grain of the 3D print, however the superglue was strong enough to hold everything securely in place.

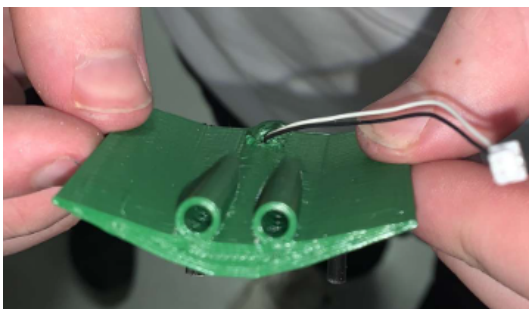


Figure 11

Motor

After drilling a hole on the front and a hole on the top of the wing joint for the wire, we pushed the motor in place and pulled the wire through the hole on top. We could then glue on the front of the wing joint to stabilise the motor.



Figures 12 & 13

Landing Gear

After drilling the hole for the wire in the main joint, we realised it was best to glue the gear on externally to avoid more fraying. We glued the frame either side of the motor mount, since the corners had the largest surface area for glueing.

We then constructed the axle, wheels and stoppers, and glued that onto the main body, ensuring the wheels could roll freely on a flat surface first.



Figure 14

The Wings

The wings were as straightforward as glueing them to the wing joint with glue.

To ensure the glue would hold, once both wings were attached, we held a wing tip and shook the glider gently to ensure they wouldn't come off during flight.



Figure 15

The 'Fuselage'

We slotted the carbon fibre rods into place and used the open ended tail joint to adjust the centre of gravity until it was where we wanted it - under the wings.

Then we cut off the remaining length and glued everything in place.

The final product is shown in figure 15.

Results



Figure 16 - Our glider in flight

Performance:

Concluding the test, our model was a success and flew smoothly around the pole several times and smoothly landed. Several comments made included how stable the plane was as well as how it was able to be controlled easily in parallel with power adjustments. Due to its very low weight, small adjustments would provide large movements since the motor's power was not reduced by the factor of weight, allowing ease of control.

Relating to material usage, we were well below our usage of 3d-printing materials, not using any of the plywood provided, also using around $\frac{1}{4}$ of the provided balsa, and using a whole carbon rod.

Comparisons:

Compared to other flyers, the relatively low weight of our glider will have meant we didn't require as much lift as others and therefore stood a better chance of taking off. The dihedral wings meant the aircraft was more stable whereas flyers without dihedral wings were not as stable and this potentially led to them not flying well. The high CL/CD ratio of 12.93 meant our aircraft has a higher aerodynamic efficiency than other groups which meant we required less power to stay at a constant height due to less drag being produced. Compared to others, our glider wings relied more on angle of attack than an aerofoil shape, which worked in our favour at lower airspeeds.

Conclusion

Summary:

Our glider's success in the test flight was due to the considerations we made in the design stage towards the conditions it would be flying in. Using aerodynamics theory and equations, we found that minimising weight was our main priority, which allowed us to have a consistent design philosophy throughout the process. We learnt about the anisotropic properties of 3D-printed components and how to factor this into the manufacturing process. Overall our glider was very efficient, both aerodynamically and in terms of materials used. Our group was effective in working together to reach a design, but also in being open to adapt and redesign at various stages.

Future Improvements:

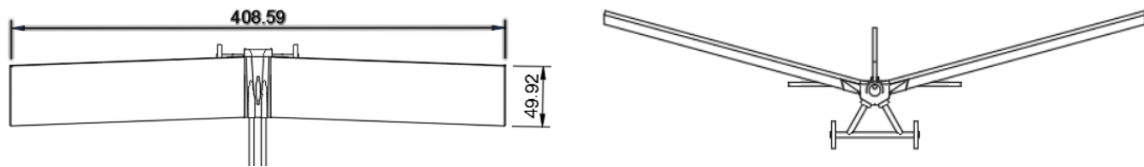
- Change the print orientation of the tail joint in order to increase strength
- Add holes to the print using teardrop shapes instead of drilling post-print
- Potentially add a second propellor to increase the speed of the aircraft
- Aerofoil the wings to increase the lift coefficient and reduce drag
- Secure the wheels on the landing gear as they were negatively cambered
- Record more data such as the motor power to improve evaluations



Figure 17 - The finished glider

Appendices

Drawings:



Calculations:

Estimated (maximum) mass: 70.374g
 Wing Span: 0.409m
 Wing Area: 0.02m²

Estimated Velocity: 5.37m/s
 Wing Chord: 0.05m
 Aspect Ratio: 8.18

At steady level flight, lift is equal to the weight:

$$L = mg$$

$$L = 0.07037 \times 9.81$$

$$L = 0.69\text{N}$$

To calculate the lift coefficient at steady level flight,

$$C_L = (2 \cdot L) / (\rho \cdot S \cdot V^2)$$

$$C_L = (2 \cdot 0.69) / (1.225 \cdot 0.02 \cdot 5.37^2)$$

$$C_L = 1.953$$

Oswald efficiency with taper ratio $\Lambda=1$:

$$e = 1 / (1 + (\Lambda/AR)^2)$$

$$e = 1 / (1 + (1/8.18)^2)$$

$$e = 0.985$$

Estimated induced drag:

$$D_i = CL^2 / (\pi \cdot AR \cdot e)$$

$$D_i = 1.953^2 / (3.14 \cdot 8.18 \cdot 0.985)$$

$$D_i = 0.151\text{N}$$

Assuming $C_{Di} \gg C_{D0}$ at non-zero lift conditions (due to low speeds and reasonably high AR), the drag and hence the thrust is:

$$T = D = L \cdot (C_{Di}/C_L) \text{ or } 0.5 \cdot C_{Di} \cdot \rho \cdot S \cdot V^2$$

$$T = D = 0.5 \cdot 0.151 \cdot 1.225 \cdot 0.02 \cdot 5.37^2 \text{ or } 0.69 \cdot (0.151/1.953)$$

$$T = D = 0.0533\text{N}$$

Hence C_L/C_D (or L/D) can be calculated:

$$C_L/C_D \text{ or } L/D = 12.93$$